

Prem Mallappa

Systems Software Architect • Embedded Systems • Virtualization

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Seasoned Software Architect with extensive expertise in designing and developing high-performance embedded systems, system software, Linux kernel drivers, and virtualization technologies. Proficient in a wide range of programming and scripting languages, with a proven track record of delivering robust, scalable solutions in complex technical environments.

A highly analytical and collaborative team player, recognized for strong problem-solving abilities, logical thinking, and a passion for mastering emerging technologies. Committed to driving innovation and excellence through clean architecture, optimized code, and best-in-class engineering practices.

Skills

Leadership	22+ years of multidisciplinary team and project leadership experience; software architecture and technical direction across systems and platform teams.
Communications	Clear written and spoken communication for architecture reviews, design discussions, mentoring, and cross-functional execution.
Technology	Broad and deep IT expertise across embedded systems, system software, cloud/security, virtualization , and software services development.
Computer Security	Cryptography , virtualization and cloud computing security architecture, secure software development , risk management, compliance, intrusion detection, and prevention.
Programming Languages	C, C++ , Assembly (x86, ARM, MIPS), Rust, Python, Go, BASH
Operating Systems	Linux , FreeBSD, Symbian
RTOS	VxWorks, QNX , seL4, Zephyr, ThreadX
Architecture	x86, ARM v6/v7/v8, MIPS64 , PowerPC, RISC-V
Development Tools	GCC, Clang, Git, CMake , Make, GDB, Perf, VTune, ARM Development Tools, Keil, JTAG, Docker, QEMU
Systems & Virtualization	Linux kernel development, device drivers , hypervisors, IOMMU/SMMU , pre-silicon validation, boot flows, and low-level firmware bring-up.
Performance & Acceleration	SIMD/vectorization, performance optimization , math and crypto libraries, SmartNIC/DPU offload , NVMe-oF, RDMA, and hardware-assisted data paths.
Specialized Skills	Embedded systems, system software, cryptography , filesystems, performance engineering, packet/data-path acceleration, and platform enablement.

Education

Masters (Computer Science)

BITS Pilani

Pilani, Rajasthan, India

2016

Bachelors (Computer Science)

Adichunchangiri Institute of Technology (AIT)

Chikkmagalur, Karnataka, India

2002

Affiliated: Visvesvaraya Technological University (VTU)

Open Source

QEMU [\(Link\)](#)

2014-2016

- SMMUv3 (IOMMU) emulation support for ARMv8
- Designed and implemented SMMUv3 model merged into mainline
- Added support for Stage1, Stage2, and nested virtualization
- Implemented command queue processing and page table walk

Linux Kernel (IOMMU/SMMUv3)

2015-2016

- Contributed to IOMMU/SMMUv3 driver development ([Fix GERRORN register write](#)) ([Fix STE.VALID check in Vulcan](#)) ([Workaround for STE.VALID](#))
- Collaborated with ARM on upstream SMMUv3 driver implementation

Linux Kernel (Kexec/Kdump) [\(Link\)](#)

2011-2013

- MIPS Kexec/Kdump port for Cavium Octeon platforms
- Developed MIPS64 port of Kexec and Kdump (merged upstream)
- Fixed critical bugs in KEXEC for Cavium Octeon platforms

GLIBC [\(Link\)](#)

2018-2020

- Performance optimizations for AMD processors
- Fixed memcpy behaviour on AMD processors
- Optimized string functions for x86₆₄ architecture
- Performance improvements for memory operations

AMD LibM [\(Link\)](#)

2018-2025

- Open source math library for AMD processors
- Core contributor to open sourcing AMD Math Library
- Optimized transcendental functions (exp, log, pow, trig)
- Implemented SIMD/FMA optimizations for vector operations

Experience

AMD Ltd.

2018 – Present

Bengaluru, India

PRINCIPAL ENGINEER

Worked in multiple teams within AMD including **AMD LibM**, **AOCL Cryptography**, **BSP**, and **NVMe** in **Pensando** BU. Every project was different, requiring quick learning and adaptation to new domains. The role also involved **leading a team** of four-to-six engineers in various projects. I was also part of the leadership team where I mentored engineers from across AOCL and Pensando BU.

SR. MEMBER OF TECHNICAL STAFF

Worked on **Model0** which was **pre-silicon validation** of next generation AMD cores, including Zen2 and Zen3. The role involved working with hardware team to bring up bare-minimum processor configuration with just core and memory controller, and run Linux, stressors and benchmarks. This involved working on low-level firmware, bootloaders, Linux kernel and performance analysis tools.

Broadcom Ltd.

2014 – 2016

Bengaluru, India

PRINCIPAL ENGINEER

Architected foundational software stack for Broadcom Vulcan, a multicore-multithreaded ARMv8 64-bit server processor. Engineered complete **SMMUv3** (System Memory Management Unit) driver infrastructure supporting advanced IOMMU virtualization. Developed and contributed an **SMMUv3 emulation model** to QEMU mainline, enabling early platform validation and ecosystem enablement. Implemented comprehensive virtualization features including command queue processing, STE/CD descriptor parsing, multi-level page table walks, and stage-1/stage-2 address translation.

Cavium India Pvt. Ltd.

2011 – 2013

Bengaluru, India

TECH LEAD

Led low-level software development for Cavium Octeon III (MIPS64) multicore network processors, delivering stable platform firmware for networking customers. Resolved a critical KEXEC kernel bug and authored the full **MIPS64 port** of the Kexec/Kdump crash dump infrastructure, ultimately merged into Linux mainline. Engineered a bare-metal core dump generation framework with a host-resident daemon that enabled detailed post-mortem GDB analysis during bring-up. Designed and implemented the **CavHv hypervisor** for MIPS64 architecture using hardware virtualization extensions to validate next-generation features.

ARM Ltd.

2005 – 2009

Bengaluru, India

SR. DEVELOPMENT ENGINEER

Executed comprehensive **OS porting initiatives** for emerging ARM architectures including ARM1176JZFS with TrustZone security extensions and the Cortex-A8 application processor. Developed a production-grade touchscreen driver for Symbian OS along with a Python-based automated validation framework used across multiple programs. Designed a precision interrupt latency measurement infrastructure that quantified TrustZone secure-world context switching overhead for silicon partners. Ported the **L4Ka::Pistachio microkernel** to ARM architecture, enabling virtualization research and multiple proof-of-concept implementations.

Sasken Communication Technologies Ltd.

2004 – 2005

Bengaluru, India

SR. SOFTWARE ENGINEER

Architected and maintained the **Extended File System (EFS)** for VxWorks-based UMTS base station infrastructure, balancing throughput with resiliency requirements. Engineered a reset-resilient filesystem with **advanced wear-leveling algorithms** so data integrity held across unexpected power cycles in the field. Designed a flash-optimized flat file tree structure that minimized write amplification and significantly extended device longevity.

Global Edge Software Ltd.

2003 – 2004

Bengaluru, India

SOFTWARE ENGINEER

Designed and implemented a **comprehensive SDIO host controller driver** for Linux on Intel StrongARM embedded platforms used in consumer devices. Engineered a high-performance 4-bit SDIO driver for Marvell 802.11g WiFi chipsets that consistently achieved maximum theoretical throughput. Optimized **DMA transfers** and interrupt handling paths to minimize latency and maximize wire-less data transfer efficiency under heavy load.

Short Stints

VSPL Ltd.

Software Architect

Nov. 2016 – Jan. 2017

Bengaluru, India

Cisco Ltd.

Software Engineer

Oct. 2010 – Apr. 2011

Bengaluru, India

B-Labs, London UK

Sr. Engineer - Contractor

Nov. 2009 – Sep. 2011

Bengaluru, India

Harman International

Engineer

Jul. 2009 – Nov. 2009

Bengaluru, India